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Innovation in Sea-Sky Line Detection: Fusion of ConvNeXt Classification, HED, and Canny Edge Detection

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Abstract

In the maritime context, the delineation of the sea-sky line, serving as a critical demarcation between the maritime and non-maritime domains, assumes paramount significance for target detection and the preservation of situational awareness. Despite this, the identification of the sea-sky line poses significant difficulties, mainly due to obstacles present in complex marine settings and the fluctuations in lighting conditions. Recognizing the pivotal role of sea-sky line detection in the realm of maritime surveillance, this paper introduces a methodological approach aimed at mitigating these challenges. The proposed methodology strategically amalgamates edge detection and deep learning classification tasks, leveraging both HED and Canny edge detection methodologies to ascertain image edges. This synergistic application not only mitigates the risk of edge loss pertaining to the sea-sky line but also diminishes the computational complexity in subsequent stages. Subsequent to edge detection, an enhanced ConvNeXt classification network is employed to discern the classification of individual edge pixel points with respect to their association with the sea-sky line. The model demonstrates noteworthy proficiency in the classification of sea-sky line edge pixels. In the final phase, the rectified sea-sky line edge pixels are fitted to the sea-sky lines utilizing the RANSAC algorithm. The efficacy of the proposed methodology is rigorously validated using the extensive Singapore Maritime Dataset. Our proposed approach has demonstrated superior accuracy through a comparative analysis with existing algorithms, highlighting its effectiveness in tackling the inherent challenges of sea-sky line detection in the maritime domain.

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CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision.

Keywords

sea-sky line detection, deep learning, edge detection, edge classification, maritime image

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1 Introduction

Target detection in marine environment monitoring, especially the accurate identification of ships, the most common target at sea, is of important significance. Identifying and tracking ships in the ocean plays a key role in the maintenance of smooth ocean shipping lanes as well as effective maritime traffic management[1]. While traditional radar has its uses in detecting marine targets, its effectiveness is influenced by various factors such as the target's shape, size, and surface properties[2]. So, we need to explore other methods to compensate for the shortcomings of radar.

Cameras are extensively employed for their capacity to capture high-resolution images. Statistical evidence suggests that ship targets frequently manifest in proximity to sea-sky lines. So, by precisely identifying the sea-sky line before initiating ship target detection, we can streamline the detection area. Hence, the detection of the sea-sky line significantly impacts the efficacy of subsequent processes, including intelligent automatic navigation, situational awareness, dynamic localization, obstacle identification, and target tracking[3]. So, accurate acquisition of sea-sky line information is crucial for all aspects of marine navigation. Traditionally, the sea-sky line is defined as a straight line that separates the expanse of the sea from the airspace above it. This particular type of detection is often treated as a special case of target detection[4]. However,

the detection of sea-sky lines faces great challenges in some cases, such as the presence of background interference and illumination variations^[5]. More effective methods are required to accurately capture the edges of the sea-sky line and to mitigate the influence of other factors present in the marine environment on the detection of sea-sky line edges. Although traditional computer vision techniques have applications in dealing with this problem, they tend to be very sensitive to noise, not adaptive to lighting changes, and susceptible to problems. It is imperative to employ computer vision methodologies that can enhance both the precision and stability associated with the detection of edges along the sea-sky line.

Convolutional Neural Networks (CNNs) showcase a remarkable capacity to grasp intricate abstract patterns, resulting in enhanced feature extraction and characterization. At present, the predominant emphasis of deep learning-based algorithms for detecting the sea-sky line lies in the realm of edge detection^[6]. Our objective is to harness the full potential of advanced CNN architectures to effectively overcome the challenges inherent in intricate marine environments^[7]. In marine scenarios, occlusion of ships and islands as well as sea waves greatly affect the accuracy of edge detection. Effectively addressing the challenge of boosting edge detection accuracy in intricate settings through the application of deep learning techniques remains an imperative concern. To address this challenge, we use the feature learning capability of deep learning models to model the edge detection method combined with the deep learning for the complex features of sea-sky lines. With the help of deep learning techniques to address the challenges in the marine environment, including occlusion and waves, our goal is to attain a higher level of precision and stability in the detection of the sea-sky line.

This paper aims to introduce a sea-sky line detection method based on deep learning, with the objective of enhancing both the accuracy and robustness of detecting the sea-sky line. Structured into three primary steps. Initially, we integrate a deep learning edge detection algorithm with the conventional edge detection method to mitigate the impact of noisy edges, aiming to preserve a greater number of sea-sky line edges. Next CNNs is used to determine whether each edge pixel point is a sea-sky line edge. Finally the corrected edge map obtained from the classification network is used to fit the accurate sea-sky line.

In summarizing the contributions of this paper, there are two key points: first, integrating the deep learning approach for edge detection with the conventional method, we improve the efficiency of the classification network and the accuracy of detecting the sea-sky line. Second, the integration of the improved convolutional neural network model with the edge detection algorithm renders it suitable for marine environments. This integration results in a method capable of precisely detecting the edge of the sea-sky line.

The subsequent sections of the paper are structured in the following sections: section II briefly reviews the related work on sea-sky line detection for ocean scenarios. Section III outlines the proposed methodology for sea-sky line detection. Section IV discusses the various experiments conducted and the results obtained. Finally, Section V summarizes the paper.

2 Related work

Methods of detection of the sea-sky line have primarily centered around traditional methods and deep learning techniques. This section will explore the three primary categories of methodologies: edge detection based methods, region based methods, and machine learning based methods.

Regarding edge-based methods, the main applications of this method for sea-sky line detection are vertical edge detection, spatial transformation, and multi-scale edge detection. The main drawback of vertical edge detection is that it assume each image column has a sea-sky line candidate point, while this assumption breaks down in the presence of occlusions. Following this is the spatial transformation phase, Shen^[8] initially apply a low-pass Gaussian filter, capable of blurring the sea-sky line. But this method is not able to detect the sea-sky line well as the sea-sky line does not always have the highest contrast. Finally is the multi-scale edge detection, which conducts edge mapping on each median-filtered image, followed by averaging the results. However, this approach proves highly susceptible to the presence of ships in the image. As for the region-based method, it mainly includes two methods: sky detection and line segment detection. Firstly, regarding the sky detection approach, which tries to identify the sky region based on the brightness and chromaticity changes in the sky region. However, this method may encounter challenges in effectively distinguishing between the sky and the sea surface. The next method is the line segment detection-based method. Usually use a line segment detector to detect line segments as candidate sea-sky lines, merging the segments into longer ones, and selecting them by the length of the line segments. But it is sensitive to erroneous candidate segments. Concerning machine learning-based approaches, these methods can be categorized into two types. First is pixel category classification. Fefilatyevev^[9] trained an SVM, J48, and plain Bayes to categorize image pixels as sea or sky. Nonetheless, this approach may fail to detect the sea-sky line when a significant portion of it is obscured by a ship. Secondly, neural networks are used for classification. Jeong^[10] use multi-scale edge detection, the obtained edges are intercepted pixel by pixel in the original image of the image block. A neural network is employed to classify image blocks, to determine whether the pixel is a sea-sky line. But due to the median filter in multi-scale edge detection, it leads to easy loss of sea-sky line edges, as well as each pixel-by-pixel judgment leads to redundant computation of image blocks. Deep learning-based methods still face some challenges such as accuracy in complex marine environments, low light and different weather conditions and the training datasets for deep learning models are usually limited by the diversity of the included scenes.

3 Methodology

This chapter proposes a method for detecting the sea-sky line in marine environments. The approach begins with edge detection using the HED algorithm, supplemented by Canny edge detection and skeletonization, aiming to identify edges while minimizing the influence of ships, islands, and waves, and preserving the sea-sky line edges. Then a deep learning network ConvNeXt is modified for the classification of the sea-sky line edges, according to the results of edge detection, traverse all the edge pixel points, take one every

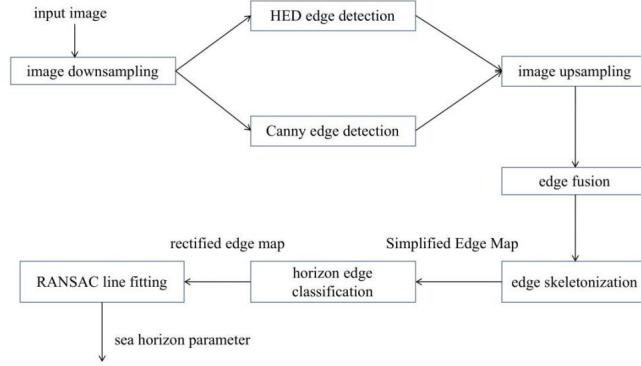


Figure 1: Flowchart of the proposed method.

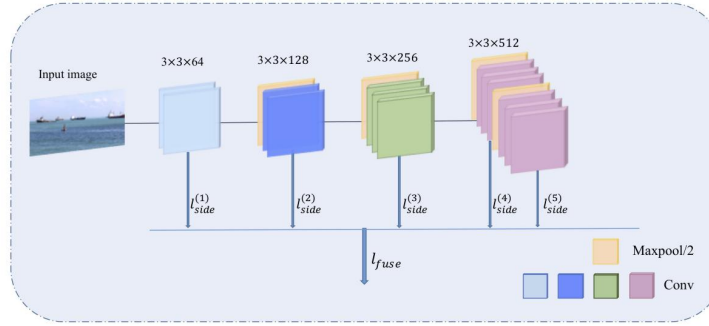


Figure 2: The structure of HED.

seven edge pixel points, take this edge pixel point as the center, and intercept the 32×32 size image in the original image, and then use the classification network for classification to determine whether the pixel point is a sea-sky line or not. Subsequently, the edge map is processed to identify pixels corresponding to the sea-sky lines, resulting in the rectified sea-sky line edge map. Ultimately, the RANSAC straight line fitting method is employed to fit the accurate sea-sky line based on the rectified sea-sky line edge map. The detailed procedure is illustrated in Figure 1.

3.1 Step1: sea-sky line edge detection

In this paper, we use HED^[11] combined with Canny edge detection algorithm to preliminarily obtain the edge of the sea-sky line. Figure 2 shows the HED network structure.

The HED algorithm, through its multi-branch convolutional network structure, is able to capture the edge information at different scales, thus extracting rich structures and contours in marine images. This multi-scale feature extraction capability provides an

advantage in dealing with marine targets of different sizes, and also allows HED to maintain a robust performance in situations with large illumination variations. However, since HED uses CNNs for edge detection, deep features may lead to thick edges detected because they focus on the overall shape and structure of the object rather than pixel-level details. In this study, we combine the HED edge detection and Canny edge detection algorithms, fusing the HED detection results with the Canny edge detection results to refine the HED edge detection results, and Equation 1 shows the process of combining the HED and Canny edge detection maps. Where E_{HED} represents the edge map, E_{Canny} represents the edge map, and "&" denotes taking the same pixels from two edge maps. E represents the fusion edge map. This approach has the capability to preserve a higher number of sea-sky line edges by HED and avoid the loss of sea-sky line edges due to manual threshold setting using Canny algorithm.

$$E = E_{HED} \& E_{Canny} \quad (1)$$



Figure 3: The representation of edge detection. Respectively, the input image, the result of HED detection, the result of Canny detection, the fusion edge and the result of skeletonization algorithm.

The edges are then further refined using a skeletonization algorithm, which is able to extract the main skeleton or central axis of an object or edge from a binary image, refining the edges to a width of 1 pixel, which reduces the workload for subsequent edge classification. Figure 3 shows the schematic representation of edge detection.

3.2 Step2: sea-sky line edge classification

There are many noise edges similar to sea-sky line in the complex Marine image. A methodology to differentiate the sea-sky line from other disruptive noise edges is virtual. In this paper, a deep learning network is applied to perform pixel-by-pixel classification of the edge map acquired through edge detection. This classification can distinguish the sea-sky line from other intrusive noisy edges. The convolutional layer extracts image features at different scales and layers, while automatically learn feature representations suitable for edge classification of sea-sky lines.

Since the sea-sky line, as a straight line dividing the water surface and the sky, its color feature is different from other noisy edge regions. The color feature can be used to identify whether a pixel belongs to the sea-sky line. So, we adopt to take one pixel every seven in the edge map in the position of the original map to intercept an image block of 32×32 size, and feed it into the neural network for classification. Image blocks are classified using an improved ConvNeXt. ConvNeXt^[12] introduces a spatial pyramid pooling layer, which performs pooling operations on features at different scales to capture contextual information at various scales. And uses a global average pooling layer in the final classification layer rather than the traditional fully connected layer, which transforms the feature map into fixed-size vectors. The introduction of multiple parallel convolutional branches allows features to be extracted at different scales. These features can be integrated by cascading or fusion to capture multi-scale information and improve the network's ability to recognize targets at different scales.

Considering the small size of the image block and the limited dataset, the original ConvNeXt which uses a convolution of 4×4 for the stem layer and a step size of 4 with the AdamW optimization algorithm did not achieve good results. So in this study, the convolution of the stem layer is modified to 1×1 with a step size of 1, and the SGD + momentum optimization algorithm is used instead

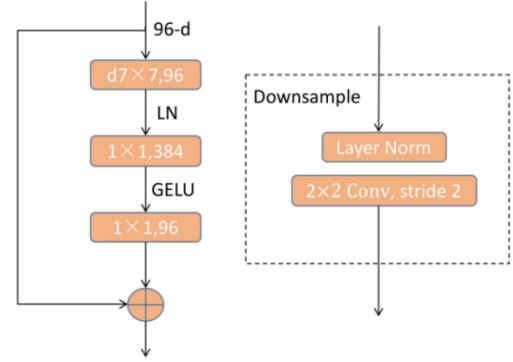


Figure 4: The elements of ConvNeXt, block of ConvNeXt and process of downsampling, respectively.

of the AdamW, and the results on the validation set are improved. Figure 4 depicts the components of the ConvNeXt network. Figure 5 shows the structure of the ConvNeXt network used in this paper.

The rectified sea-sky line edge map is obtained after determining whether a pixel point is a sea-sky line edge by the classification network, which helps to improve the accuracy of sea-sky line fitting, and Figure 6 shows the rectified sea-sky line edge map.

3.3 Step3: sea-sky line fitting

In this paper, we use the RANSAC linear fitting algorithm^[13]. The RANSAC algorithm performs well in dealing with datasets containing noise and outliers. Compared to the Hough transform and linear curve fitting, RANSAC can better handle the task of fitting the sea-sky line. The method performs the fitting by randomly sampling some data points, calculating the fitting error, and marking the data points with error less than a given threshold as interior points. Then, the fitting model is continuously optimized through an iterative process to mitigate the effect of anomalous noise edges in the ocean scene and obtain more reliable straight line fitting results.

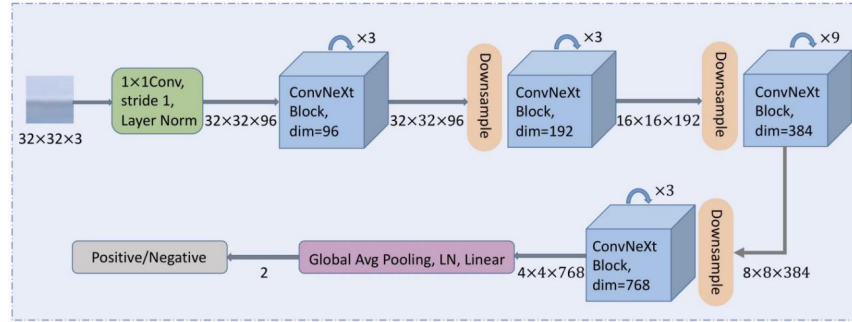


Figure 5: The structure of ConvNeXt.

composition is 6781 negative samples and 4029 positive samples. An example of making samples is shown in Figure 8.

4.2 Comparison of various models for classification of sea-sky lines

This section provides a comparison of the performance of seven deep learning models in a sea-sky line classification task. All models are built using Tensorflow, the optimization algorithm uses stochastic gradient descent or AdamW, 35 epochs, the learning rate is set to 0.001, and the inputs are all 32x32 image blocks except for AlexNet which is fed with 224x224 image blocks. Table 1 shows the accuracy of each network on both the training and validation sets.

As can be seen from the results, GoogleNet, ConvNeXt-T and the Improved ConvNeXt achieved good results on the training set while the improved ConvNeXt has achieved the best result on the test set.

4.3 Performance of sea-sky line detection and comparison with other methods

This subsection specifically compares the performance of the methods used in this paper to previous methods. The results of MuScoWERT are from Prasad^[15] and the results of MSCM-LiFe, LHSL, IntV, Hough, and Jeong's are from Chiyeon Jeong^[10]. MuScoWERT and MSCM-LiFe use the multi-scale method. They are better than those use single scale method(LHSL, IntV, Hough). However, all of them are unstable when ships exist in the maritime image. Jeong's improved a lot for the use of CNNs. However, the use of multi-scale convolutions with Canny to detect edge is difficult to adjust the parameters, while simple design of the model still leads to some mistakes of classification. Our method applies a more intricate model to further improve the accuracy. The comparison with previous methods is shown in Table 2 and Table 3. In terms of the error about Y, our method outperforms the previous best method by 0.09, 0.12, and 0.86 at the 25th percentile, 50th percentile, and 95th percentile, respectively. While by 0.06, 0.63, and 0.89 at the

Figure 6: The rectified sea-sky edge map.

4 Results and discussion

4.1 Dataset

In many prior studies, researchers have utilized datasets that they personally collected, but this approach does not guarantee the quality of the dataset and there are no metrics to measure the accuracy of the final detected sea-sky lines. In this paper, we use the publicly available Singapore Maritime dataset (SMD)^[14]. It contains a variety of marine scenes covering diverse weather and lighting conditions, with ships and islands in the videos. Singapore Maritime is divided into two datasets, and the graphs are labeled with the distance of the midpoint of the sea-sky line from the top, Y, as well as the slope of the sea-sky line, α . Figure 7 is the sample image of SMD.

In this paper, the SMD is used to produce a dataset by intercepting image blocks of size 32x32 in the figure manually. The image blocks containing the sea-sky line are taken as positive samples and the others as negative samples. The intercepted positive and negative samples are used to train and test the deep learning network. Meanwhile, the results of edge detection will be greatly affected due to the sea surface clutter. So we add a certain amount of sea surface clutter in the negative samples, and experiments have proved that it can well avoid the influence of sea surface clutter on edge detection and improve the efficiency when classifying. The specific sample



Figure 7: The sample image of SMD. The sample image of onboard dataset and onshore dataset.

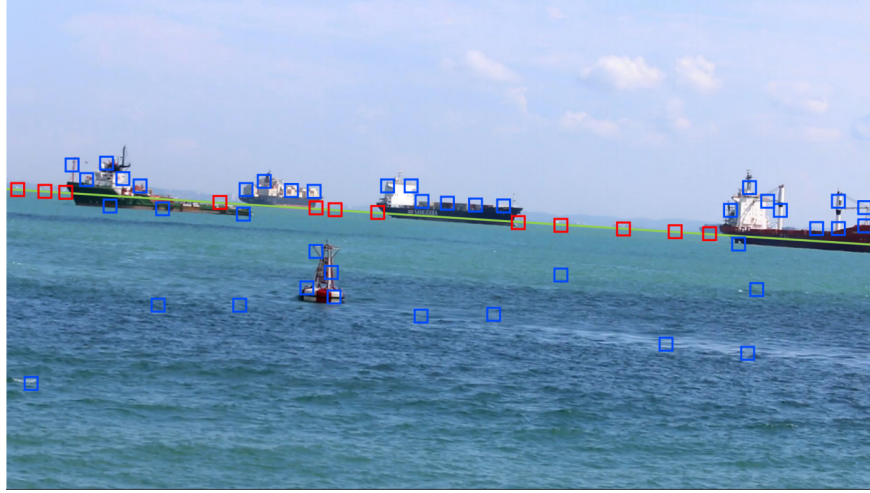


Figure 8: Sample of manufactured dataset for CNN. The red rectangles denote positive and the blue rectangles denote negative. The green line denotes the groundtruth of sea-sky line.

Table 1: The accuracy of each network on both the training and test sets.

	train_accuracy(%)	test_accuracy(%)
AlexNet	97.97	96.88
VGG16	95.90	95.95
ResNet50	96.39	97.10
GoogleNet	100	98.74
ShuffleNet	99.7	98.1
ConvNeXt-T	100	97.9
Improved ConvNeXt	100	99.4

onboard dataset, respectively. And the error about α , our method outperforms the previous best method by 0.01, 0, and 0.1 at the 25th percentile, 50th percentile, and 95th percentile, respectively. While by 0, 0.01, and 0.12 at the onshore dataset, respectively.

The sample frame test results for various types of sceners in the SMD dataset are shown in Figure 9. It shows the detection results with the method on the onboard and onshore dataset.

5 Conclusion

In this paper, we propose to combine HED with Canny edge detection algorithm and skeletonization algorithm, and classify the edges by the improved ConvNeXt network to get the sea-sky line edges, which makes it possible to detect accurate sea-sky lines through the images. Integrating HED with Canny edge detection algorithm and skeletonization algorithm is able to get finer edges, which reduces the computation of subsequent classification network. We modified the stem layer and optimizer of the ConvNeXt network to make

Table 2: The comparison result of Y, α on the onboard dataset of SMD.

	$ Y - Y_{GT} (\text{pixels})$			$ \alpha - \alpha_{GT} (^{\circ})$		
	25th percentile	50th percentile	95th percentile	25th percentile	50th percentile	95th percentile
MuScoWERT	0.54	1.49	8.17	0.06	0.25	0.88
MSCM-LiFe	1.16	2.84	505.78	0.17	0.38	5.50
LHSL	13.78	25.65	507.92	0.88	1.37	6.52
FGSL	5.28	10.85	581.44	0.67	1.00	3.88
IntV	13.36	24.89	498.17	0.87	1.35	6.12
Hough	2.27	221.67	520.34	0.25	1.00	4.57
Jeong's	0.53	1.21	3.79	0.04	0.10	0.38
Ours	0.44	1.09	2.93	0.03	0.10	0.28

Table 3: The comparison result of Y, α on the onboard dataset of SMD.

	$ Y - Y_{GT} (\text{pixels})$			$ \alpha - \alpha_{GT} (^{\circ})$		
	25th percentile	50th percentile	95th percentile	25th percentile	50th percentile	95th percentile
MuScoWERT	1.14	2.63	11.41	0.14	0.21	1.07
MSCM-LiFe	1.63	3.88	81.59	0.11	0.18	1.14
LHSL	14.96	27.92	109.00	0.75	1.03	3.86
FGSL	5.88	11.53	64.70	0.75	1.00	2.87
IntV	2.08	5.82	39.89	0.14	0.52	5.37
Hough	3.12	165.02	460.24	0.14	0.36	3.80
Jeong's	0.85	1.71	10.89	0.03	0.12	0.82
Ours	0.79	1.08	10.00	0.03	0.11	0.70

**Figure 9: Examples of sea-sky line detection results.**

it more suitable for small-sized images in marine environments and improve the classification accuracy. The effectiveness of our method is validated on SMD.

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